

Bi-Weekly Status Report 2

CodeTrail – Real-Time Collaborative Code Editor with Contribution Tracking

Course: WS & SOA (IT644) | Dhirubhai Ambani University (Formerly DA-IICT)

Report Period: 18/08/2026 – 31/08/2026 (Weeks 4–5) | Date Submitted: 01/09/2026

Prepared By: Anistina Dsouza on behalf of CodeTrail Team

1. Project Status Summary

Sprint / Milestone: SCRUM Sprint 2 (Bi-Weekly 2) – Module 2 Delivery, Requirements & System Design

Overall Completion: 25% (2 of 8 functional modules delivered On Track)

Indicator	Current Status	Previous Status	Trend	Overall
Overall Project	On Track	On Track	→	On Track
Schedule	On Track	On Track	→	On Track
Scope	On Track	On Track	→	On Track
Quality	On Track	On Track	→	On Track
Team Contribution	On Track	On Track	→	On Track
Dependencies	On Track	On Track	→	On Track

Summary Notes:

In accordance with our project roadmap to submit one complete module deliverable every bi-weekly cycle, the CodeTrail development team focused this second bi-weekly period (18/08/2026 – 31/08/2026, Weeks 4–5) on completing and submitting **Module 2 (Workspace Creation & Collaboration Management)**, alongside the formal completion of the foundational engineering design documents:

- **Module 2 (Workspace Creation & Management) – Target Module Delivered:** Fully implemented backend REST APIs (/api/workspaces CRUD, cryptographic 8-character invite code generation, participant role permissions by Anistina Dsouza) and comprehensive frontend modal UI with dual-tab creation & join-via-code workflows, multi-language boilerplate initialization (JavaScript, Python, C++, Java, Go, Rust), auto-jump navigation to joined rooms, and responsive grid layout on the live HomePage dashboard (by Anistina Dsouza & Shah Rishika).
- **System Design Document (SDD):** Completed and formally documented by **Meet Sheth**, detailing the complete End-to-End System Architecture (Presentation layer, Express REST API gateway, Socket.IO real-time server, MongoDB Atlas with Mongoose ODM, and external Piston API cloud runner), Database Schema / ER Diagram (8 core entities), UML Class Diagram, Main System Sequence Diagram, REST API contracts table, and UI/UX design mockups.
- **Requirements Specification Document (BRD / RSD):** Authored collaboratively by **Rishika Shah** and **Mahi Kansara**, and formally submitted to the repository as RequirementsSpecificationResearchCodeTrail.pdf (PR #12). Rishika authored all 25 Functional Requirements (FR-01 to FR-25), Detailed User Stories (US-01 to US-17), Acceptance Criteria Matrix (AC-01 to AC-41), Data Requirements, and External Interfaces. Mahi authored all 10 Non-Functional Requirements (NFR-01 to NFR-10), the Use Case Model and Diagram (UC-01 to UC-05), Interface Requirements, Initial UI/UX Wireframe specifications, Constraints, Assumptions, and Dependencies.
- **User Profile & Account Management:** Completed profile editing, personal info management, avatar customization, and secure password change controllers.

2. Key Accomplishments in Two Weeks

- **Module 2 Delivery (Target Bi-Weekly Deliverable):** Successfully completed, merged, and integrated into main (commits 3667336e, 069f2614, f16eff1b, 3c757aeb). Covers backend REST endpoints, Mongoose schema, cryptographic 8-character invite code generation, participant permissions, dual-tab WorkspaceModal.jsx (Create & Join via Code), auto-jump routing, and dashboard polish.
- **Formal Submission of Requirements Specification Document (RSD):** Formally uploaded and archived in the repository as RequirementsSpecificationResearchCodeTrail.pdf via PR #12 (commit a5f6018f). Produced by Shah Rishika and Mahi Kansara, comprising 25 Functional Requirements, 10 Non-Functional Requirements, 17 User Stories, 5 Detailed Use Cases, 41 Acceptance Criteria, and full UI Wireframe specifications.
- **Submission of System Design Document (SDD):** Produced by Meet Sheth, establishing Client-Server Architecture, MongoDB ER Schema, UML Class Diagram, Main Sequence Diagram, and OpenAPI REST API contracts.
- **Jira Backlog Organization:** Populated and categorized 395 work items under parent task CT-325 mapped to Epics and User Stories.
- **User Profile & Security Settings:** Implemented protected profile endpoints (/api/auth/profile, /api/auth/change-password) and frontend profile components (PR #8, PR #11).
- **Dashboard Polish & Quick Actions:** Enhanced HomePage.jsx with quick "Join via Code" action, normalized language tags, active filter glow styling, dynamic backend fetching, search, and seamless room entry.

3. Planned vs. Actual Progress

Task / Deliverable	Planned	Current	Actual	Notes
	Completion	Forecast	Status	
Requirements Eng. (BRD / RSD)	Aug 24	—	Done	Authored by Rishika & Mahi; formal PDF submitted (PR #12)
System Design (SDD)	Aug 31	—	Done	Authored by Meet (SDD finalized)
Module 1: Auth & User Mgt	Aug 17	—	Done	Submitted in Bi-Weekly 1 (On Track)
Module 2: Workspace Mgt	Aug 31	—	Done	Delivered & merged to main (APIs + Modal UI, On Track)
Module 3: Collaborative Editor	Sep 14	Sep 14	In Progress	Target for Bi-Weekly 3 (Monaco & Socket.IO)
Module 4: Code Execution	Sep 28	Sep 28	Scheduled	Target for Bi-Weekly 4 (Piston API runner)

4. Project Risks, Issues, & Dependencies

Type (ID)	Description	Impact	Owner	Mitigation / Next Step
Risk (R-02)	Concurrent typing race conditions during live collaborative editing	High	Mahi Kansara / Anistina Dsouza	Implement debounced delta event broadcasting and versioned cursor synchronization over Socket.IO rooms.
Issue (I-02)	Concurrency collisions on atomic participant additions via invite codes	Medium	Manushri Thakkar / Ruchita Patadiya	Implemented atomic <code>\$addToSet</code> and unique indexing in Mongoose schema to avoid race conditions.
Dependency (D-02)	Third-party Piston API execution rate limits and latency variability	Medium	Meet Sheth / Monar Dudhat	Designed execution timeout guards, client debouncing, and local execution result caching.

5. Project Decisions

Decision	Rationale	Impact	Decided By	Date
Cryptographic 8-char Invite Codes	Simplifies room joining for hackathons without requiring email invites	Frictionless room joining and instant collaboration	Team	Aug 20
Pre-populated Starter Templates	Provides immediate syntax across JS, Python, C++, Java, Go, Rust	Instant execution readiness upon workspace creation	Anistina / Ruchita	Aug 24
Dual-Mode Workspace Modal & Quick Join	Integrates creation and code-join into a unified modal with uppercase normalization and auto-jump routing	Frictionless onboarding and instant room collaboration	Rishika / Anistina	Aug 30
Modular Tab Architecture	Separates Editor, Live Console/Terminal, Dossier, and Settings panels	Clear modularity & frontend maintainability	Maryam / Rishika	Aug 25
Monaco Editor & Socket.IO	Industry-standard VS Code engine with rich syntax highlighting and cursor positioning	Superior editing and collaborative UX	Team	Aug 28

6. Project Scope Changes

Change	Action	Reason	Schedule Impact	Approved By
None	—	Scope remains fully aligned with project proposal and approved SDD specifications.	None	—

7. Upcoming Work (Next Two Weeks' Plan – Bi-Weekly 3 Target)

- Deliver **Module 3: Real-Time Collaborative Code Editor** (Monaco Editor integration, multi-user live cursors, Socket.IO broadcast handlers).
- Implement dynamic multi-file management (add, delete, rename files within an active workspace session).
- Prepare Piston API client infrastructure in advance for **Module 4: Code Execution Engine**.
- Establish core SHA-256 cryptographic logging pipeline for tamper-evident contribution recording.

8. Team Member Contributions

Member Name (ID) & Role	Tasks Completed (Key Deliverables)	Tasks In Progress	Hours	Evidence / Artifacts
Meet Sheth 202512057	Authored System Design Document (SDD): Formulated end-to-end System Architecture Diagram (React, Express, MongoDB, Socket.IO, Piston API); designed Database Schema (ER Diagram, 8 core entities); constructed UML Class Diagram and Main Sequence Diagram; defined REST API specification table and OpenAPI contracts; created UI/UX mockups.	Subtask CT-450 (Piston API client service wrapper and sandbox execution integration) – 50% complete.	28/30	• Complete SDD Document • Architecture & ER Diagrams • UML Class & Seq Diagrams
Rishika Shah 202512041	Co-authored RSD (Functional Lead): Formulated 25 Functional Requirements (FR-01–25), 17 User Stories (US-01–17), Acceptance Criteria Matrix (AC-01–41), Data Requirements, and External Interfaces. Workspace Modal UI & Quick Join: Implemented dual-mode modal (<code>WorkspaceModal.jsx</code>) with create and join-by-code tabs, uppercase formatting, auto-focus ref, and error feedback; integrated "Join via Code" header action and auto-jump room routing in <code>HomePage.jsx</code> ; styled card grid and active filters in <code>index.css</code> (merged in 3c757aeb).	Subtask CT-455 (Collaborative editor test scenarios and validation scripts for FR-07 to FR-12) – 40% complete.	30/30	• RequirementsSpecification-ResearchCodeTrail.pdf • Commits <code>f16eff1b</code>, <code>3c757aeb</code> • <code>WorkspaceModal.jsx</code> • <code>HomePage.jsx</code>, <code>index.css</code>

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Member Name (ID) & Role	Tasks Completed (Key Deliverables)	Tasks In Progress	Hours	Evidence / Artifacts
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Mahi Kansara 202512078	Co-authored RSD (Use Case & UI Lead): Formulated 10 Non-Functional Requirements (NFR-01–10), Use Case Model & Diagram (UC-01–05), Interface Requirements, UI/UX Wireframe specs (Sections 11, 13, 14), Constraints, Assumptions, and Dependencies; built user profile frontend modules.	Subtask CT-445 (Monaco Editor setup and Socket.IO synchronization architecture) – 45% complete.	30/30	• RSD Sections 4, 5, 7, 9–11, 13–14 • <code>frontend/src/modules/profile/</code> • Profile UI components
Thakkar Manushri 202512122	Coordinated BRD, SDD, and Module 2 milestone deliverables; organized WBS task tracking on Jira (parent CT-325 with 395 work items); maintained Git repository branches, reviews, and merges (PR #12 document upload, branch merges for Module 2); designed MongoDB database schemas and indexing rules.	Subtasks CT-420 (Socket.IO room architecture) and CT-425 (Database query optimization) – 45% complete.	28/30	• Milestone coordination • PR approvals (PR #11, #12, 3c757aeb) • Jira sprint backlog (CT-325)
Anistina Dsouza 202512081	Target Module 2 Delivery Lead: Built <code>workspaceController.js</code> , <code>workspaceRoutes.js</code> , and <code>Workspace.js</code> model (CRUD, 8-character cryptographic invite codes, participant role assignments); created foundational frontend <code>WorkspaceModal.jsx</code> supporting multi-language boilerplate initialization (JS, Python, C++, Java, Go, Rust).	Subtask CT-435 (Workspace participant role permissions & member management API) – 60% complete.	30/30	• Commits 3667336e, 069f2614 • <code>workspaceController.js</code> • <code>Workspace.js</code>, <code>WorkspaceModal.jsx</code>
Ruchita Patadiya 202512070	Enhanced backend security middleware (<code>middleware/auth.js</code>) with JWT bearer token verification and error handling; implemented backend endpoints for profile retrieval/updates and secure password changes (<code>/api/auth/profile</code> , <code>/api/auth/change-password</code>); integrated user session checks.	Subtask CT-430 (Workspace authorization & role-based route guards) – 50% complete.	30/30	• Commit c21dac68 • <code>middleware/auth.js</code> • <code>authController.js</code>

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Member Name (ID) & Role	Tasks Completed (Key Deliverables)	Tasks In Progress	Hours	Evidence / Artifacts
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Shaikh Maryam 202512074	Developed responsive styling architecture and design system in <code>index.css</code> ; refined landing page and user dashboard navigation; integrated modular tabs for profile settings and workspace layout; resolved CSS specificity and layout responsiveness issues.	Subtask CT-478 (Collaborative editor layout and multi-panel split view) – 55% complete.	28/30	• <code>frontend/src/index.css</code> • <code>HomePage.jsx</code> • Dashboard layout standardizations
Dudhat Monar 202512019	Formulated comprehensive test scenario matrices and Postman API test collections for authentication, password recovery, and workspace CRUD endpoints; verified invite-code join workflows and token expiration boundaries; supported backend route validation.	Subtask CT-440 (Automated API integration test suite for workspace routes) – 40% complete.	26/30	• Postman collection test exports • API test documentation • Workspace verification logs